

AI MENTAL HEALTH

ANA – 3 (WEEK-3)

Prescribing Trends Analysis

**Dataset Used: OpenPrescribing — University of Oxford /
NHS prescribing data:**

- **Antidepressant Prescribing Dataset**
- **Anxiolytic Prescribing Dataset**

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1. Introduction

Mental health conditions such as depression and anxiety are among the most common health issues treated within primary care services. Antidepressants and anxiolytics are widely prescribed to support treatment and symptom management. Monitoring prescribing trends helps healthcare organizations understand regional demand, identify prescribing variations, and evaluate healthcare resource allocation.

This project analyses prescribing data obtained from Open Prescribing datasets at Clinical Commissioning Group (CCG) level. The analysis focuses on identifying prescribing trends, regional prescribing patterns, and prescribing cost variations for antidepressant and anxiolytic medications.

2. Objectives

The main objectives of this analysis are:

- Analyze antidepressant prescribing trends over time
- Analyze anxiolytic prescribing trends over time

- Compare prescribing volumes between medication categories
- Identify top prescribing CCG regions
- Perform regional prescribing summary analysis
- Analyze prescribing costs across regions
- Generate trend charts and regional comparison visualizations

3. Dataset Description

From OpenPrescribing — University of Oxford / NHS prescribing data

Two datasets were used

Dataset	Description
Antidepressant Prescribing Dataset	Contains prescribing information for antidepressant medications
Anxiolytic Prescribing Dataset	Contains prescribing information for anxiolytic medications

Important Fields Used

Column Name	Description
date	Prescribing date
name	CCG / region name
y_items	Number of prescription items
y_actual_cost	Actual prescribing cost

4. Tools and Technologies

Tool	Purpose
Python	Data analysis
Pandas	Data preprocessing and aggregation
Matplotlib	Data visualization
CSV Files	Dataset storage
Google Colab	Development environment

5. Methodology

5.1 Data Loading

Both prescribing datasets were imported using Pandas.

```
“#Load Datasets
```

```
antidepressant_df = pd.read_csv("/content/items for antidepressant drugs  
per.csv")
```

```
anxiolytic_df = pd.read_csv("/content/items for anxiolytics per.csv”)”
```

5.2 Data Preprocessing

The following preprocessing steps were performed:

- Converted date columns into datetime format
- Extracted year and month information
- Checked dataset structure and field consistency
- Removed unnecessary formatting issues

```
“antidepressant_df['date'] = pd.to_datetime(antidepressant_df['date'])  
anxiolytic_df['date'] = pd.to_datetime(anxiolytic_df['date'])”
```

5.3 Monthly Trend Analysis

Monthly prescribing totals were calculated using grouped aggregations.

```
anti_trend = antidepressant_df.groupby(  
    ['Year', 'Month']  
)[['y_items']].sum()
```

5.4 Comparative Trend Analysis

Antidepressant and anxiolytic prescribing trends were merged into a combined dataset to compare prescribing behaviour between both drug categories.

The comparison plot highlighted prescribing differences over time.

5.5 Top Prescribing CCG Analysis

The top prescribing CCGs were identified by aggregating prescription item counts.

```
“top_anti_ccg = antidepressant_df.groupby(  
    'name'  
)[['y_items']].sum()”
```

This helped identify regions with the highest prescribing volumes.

5.6 Regional Prescribing Summary

Regional summaries were created using:

- Total prescription items
- Total prescribing cost
- Average prescribing items

```
“regional_anti_summary = antidepressant_df.groupby('name').agg(  
    Total_Items=('y_items', 'sum'),  
    Total_Cost=('y_actual_cost', 'sum'),  
    Average_Items=('y_items', 'mean')  
)”
```

6. Data Visualization

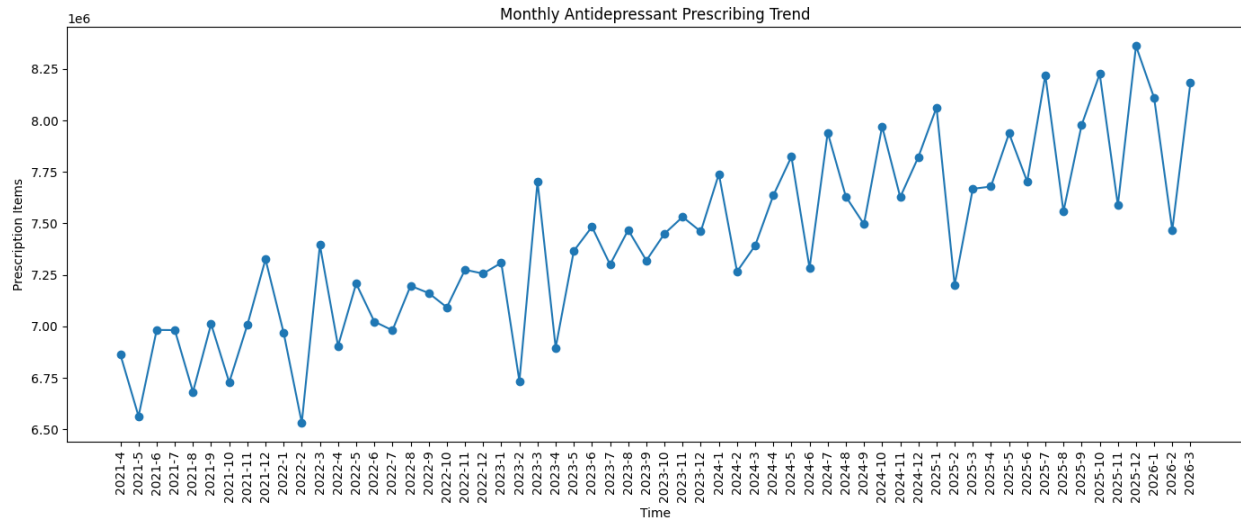
6.1 Monthly Antidepressant Prescribing Trend Chart

A line chart was generated to analyze monthly antidepressant prescribing activity over time.

Purpose

The purpose of this visualization was to:

- Identify increases or decreases in prescribing behavior
- Observe long-term prescribing patterns
- Detect seasonal variations in prescription demand
- Compare prescribing activity across months and years



The chart showed:

- Consistently high antidepressant prescribing volumes
- Gradual long-term growth in prescription activity
- Several periods with noticeable prescribing increases
- Stable prescribing demand across most months

This indicates sustained use of antidepressant medications within primary care settings.

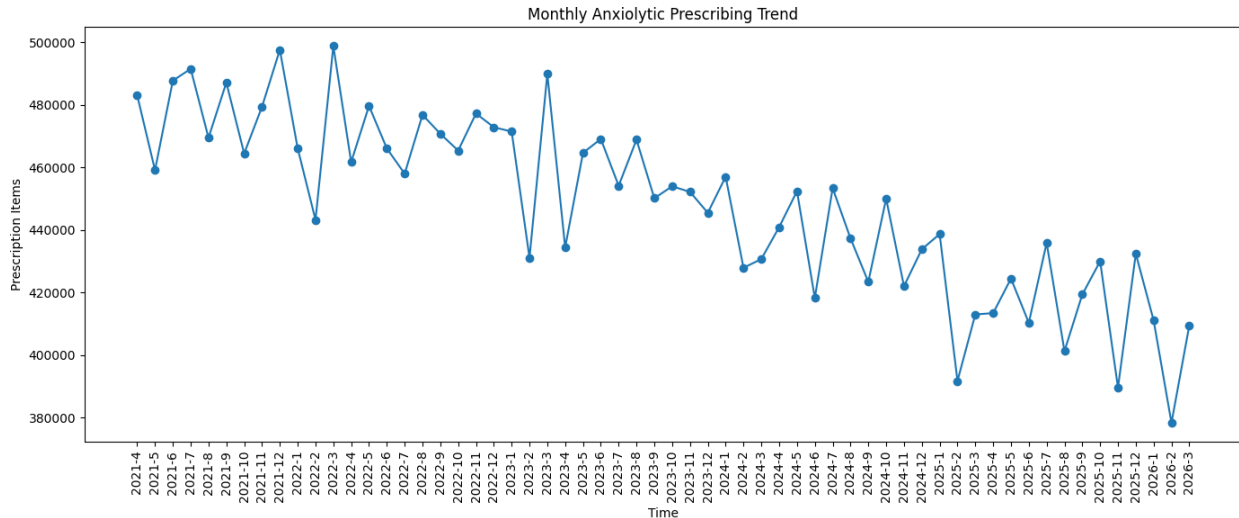
6.2 Monthly Anxiolytic Prescribing Trend Chart

A separate line chart was created for anxiolytic prescribing trends.

Purpose

This visualization helped to:

- Track anxiolytic prescribing activity over time
- Compare prescribing levels with antidepressants
- Identify fluctuations in prescription demand
- Detect periods of increased anxiety-related prescribing



The visualization indicated:

- Lower prescribing volumes compared to antidepressants
- Moderate month-to-month variation
- Stable prescribing patterns over time
- Certain months with elevated prescribing demand

The chart demonstrates continued reliance on anxiolytic medications within mental health treatment.

6.3 Combined Prescribing Trend Comparison

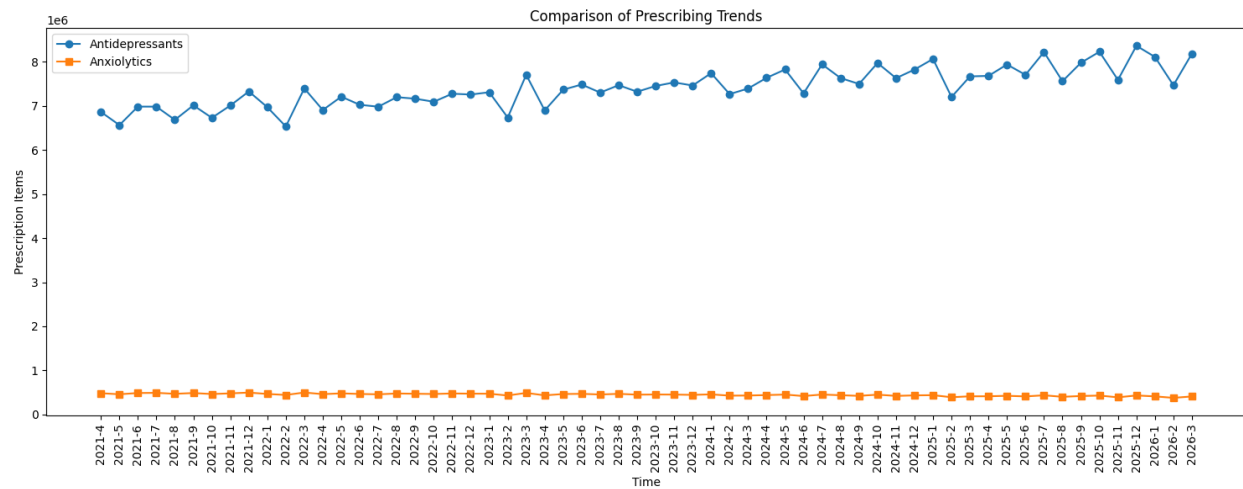
A combined trend chart was created to compare antidepressant and anxiolytic prescribing behavior on the same graph.

Purpose

This visualization was designed to:

- Compare prescribing trends between both medication groups
- Identify relative prescribing differences
- Observe whether prescribing behaviors follow similar patterns

- Improve comparative trend analysis



The combined chart showed:

- Antidepressant prescribing volumes were significantly higher
- Both medication groups followed similar overall trend patterns
- Prescribing behavior remained relatively stable
- Trend synchronization may indicate broader mental health demand patterns

This visualization provided a clearer comparative understanding of prescribing behavior.

6.4 Top Prescribing CCG Bar Charts

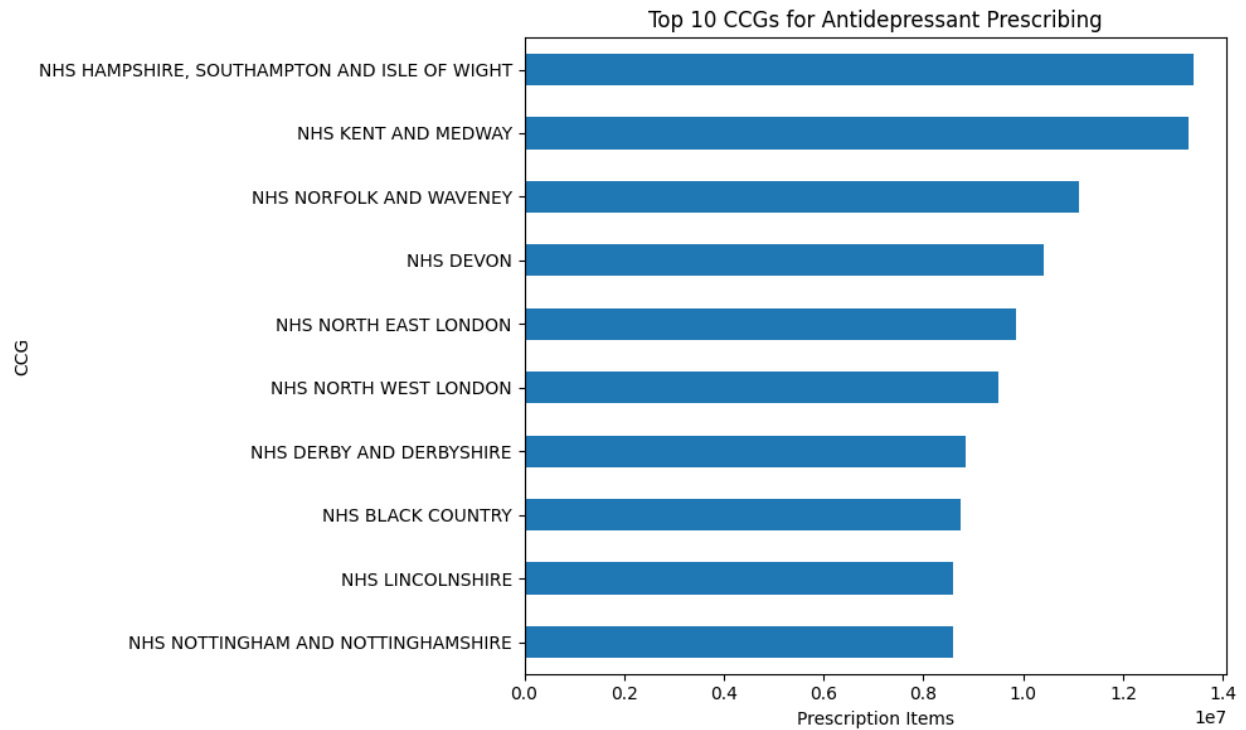
Horizontal bar charts were generated to identify the highest prescribing CCG regions.

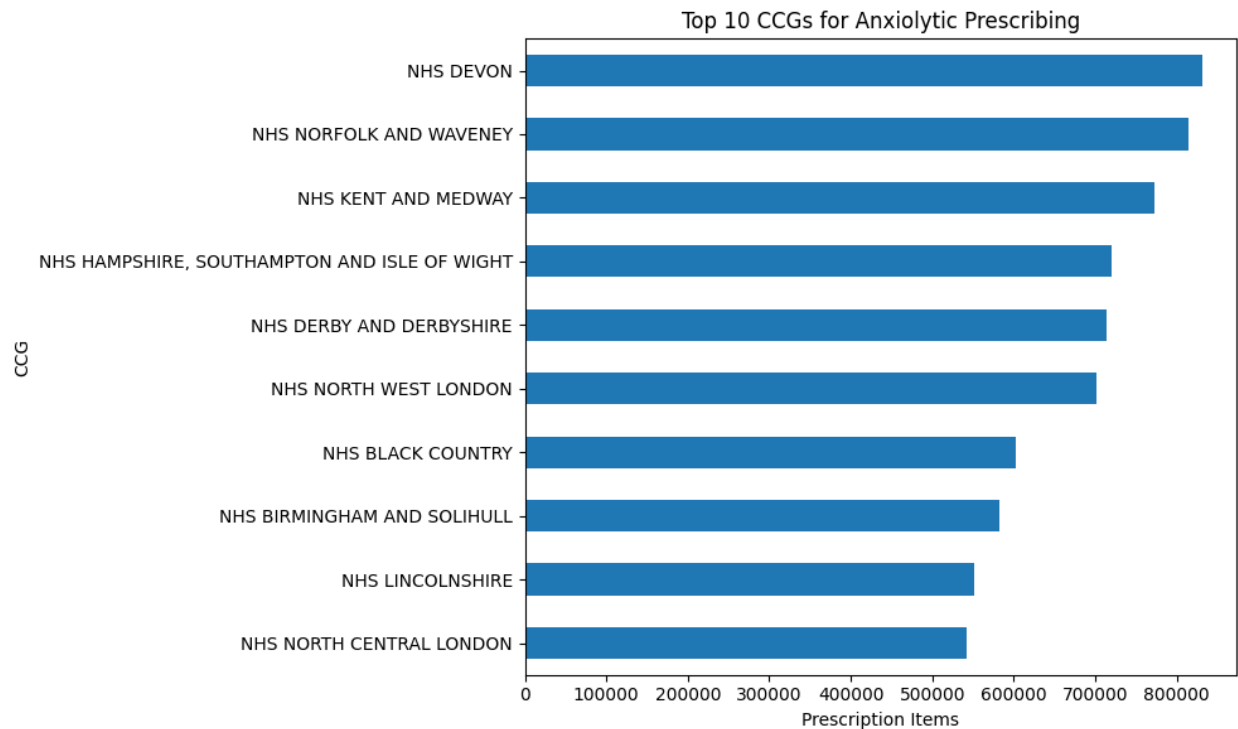
Purpose

These charts were used to:

- Identify top prescribing regions

- Compare prescribing volumes between CCGs
- Detect regional prescribing variation
- Support regional healthcare evaluation





The bar charts revealed:

- Significant differences between CCG prescribing volumes
- Several regions consistently showed high prescribing activity
- Prescribing demand varied considerably across locations

This may also reflect:

- Population size differences
- Regional healthcare demand
- Mental health prevalence variation
- Differences in prescribing practices

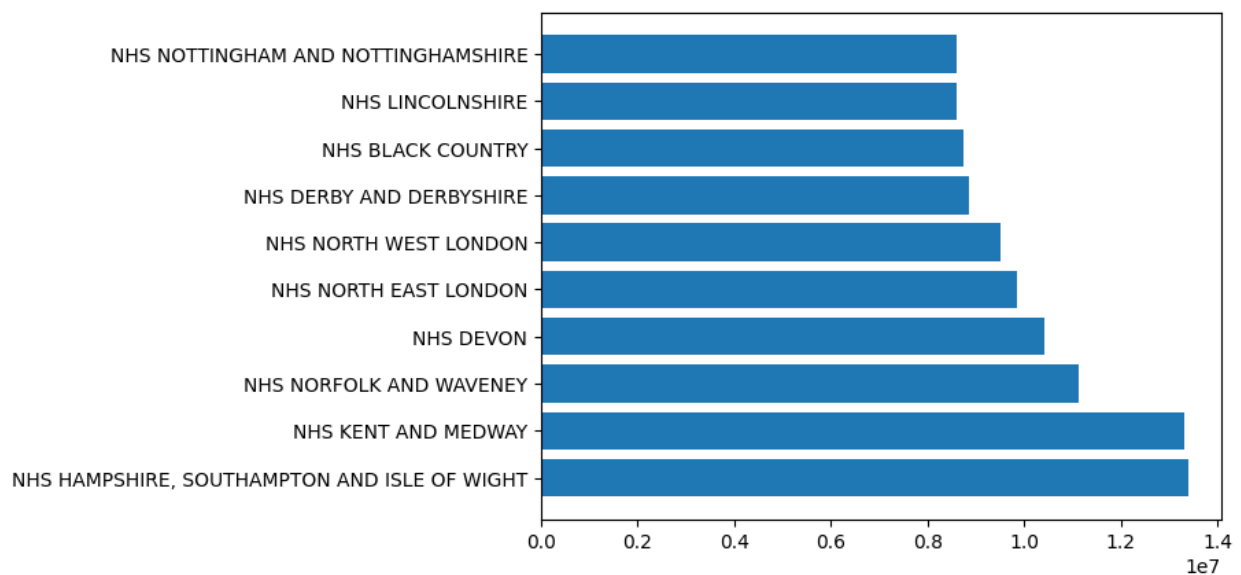
6.5 Regional Prescribing Summary Visualization

Regional summary visualizations were created using aggregated prescribing statistics.

Purpose

The regional summary charts helped:

- Compare total prescribing activity
- Analyze prescribing costs
- Evaluate average prescription behavior
- Identify high-cost prescribing regions



It highlights:

- Regional prescribing concentration
- High-cost prescribing regions
- Variations in average prescription volumes
- Uneven distribution of prescribing activity

These findings may help support NHS prescribing reviews and healthcare planning decisions.

7. Results and Findings

7.1 Antidepressant Prescribing Trends

- Antidepressant prescribing volumes were consistently high across the observed period.
- Some months showed increased prescribing activity indicating seasonal or demand-related variation.
- Prescribing levels generally remained stable with gradual long-term growth.

7.2 Anxiolytic Prescribing Trends

- Anxiolytic prescribing volumes were lower than antidepressants.
- Regional differences were observed across CCGs.
- Certain regions demonstrated significantly higher prescribing activity.

7.3 Comparative Analysis

The combined comparison chart showed:

- Antidepressants were prescribed more frequently than anxiolytics
- Prescribing patterns for both medication categories followed similar trend movements
- Regional prescribing behavior varied considerably

7.4 Regional Prescribing Summary

The regional summary analysis identified:

- High-prescribing CCG regions

- Large differences in prescribing costs
- Variations in average prescription item counts

This analysis can support healthcare planning and regional prescribing evaluation.

8. Output Files Generated

File Name	Description
Combined_Prescribing_Trends.csv	Combined monthly trend data
Regional_Prescribing_Summary.csv	Regional prescribing summary
Trend Charts	Prescribing trend visualizations

9. Conclusion

This successfully analyzed antidepressant and anxiolytic prescribing trends using Open Prescribing datasets. The analysis identified prescribing variations across CCG regions and highlighted long-term prescribing behavior.

The generated visualizations and summary datasets can support:

- Healthcare monitoring
- Regional prescribing evaluation
- NHS resource planning
- Mental health policy analysis